

Visual input trumps auditory input: teaching English intonation

Lara Portmann¹, Adrian Leemann²

¹University of Bern, ²Lancaster University

lara.portmann@ens.unibe.ch, a.leemann@lancaster.ac.uk

Abstract

Audio-visual input has long been reported to be effective for teaching second language (L2) intonation. An examination of visual-only input, however, has not been conducted yet. Using a between-subjects design, we investigated the effectiveness of auditory-only and visual-only input for teaching English intonation to speakers of Bernese Swiss German. Our results indicate that speakers who receive visual-only input improve intonation more than speakers who receive auditory-only input. This finding is in line with research which showed that audio-visual input is more effective for teaching intonation than audio-only input. Our findings are new, however, as they indicate that also visual input alone can lead to substantial improvement in attaining target intonation.

1. Introduction

Transfer errors in intonation can lead to substantial cross-cultural miscommunication (Popkova 2015; Pickering 1999), thus increasing the need for intonation teaching in second language acquisition (SLA). Several studies have shown successful use of pitch visualisation displays for teaching L2 intonation. Pitch displays, dating back to the 60s, were originally used for work with the hard of hearing (Hardison 2010) but were soon extended to SLA through the pioneering work of De Bot (1980, 1983; De Bot & Mailfert 1982; Weltens & De Bot 1984), who successfully used audio-visual input to teach intonation. Early research focused on sentence-level intonation (Hardison 2010). Levis and Pickering (2004) and Hardison (2005) have since shown that similar approaches can also be used to teach discourse-level intonation, with Hardison (2004, 2005) showing that teaching effects may be generalisable to novel sentences.

While the effects of audio-visual input for teaching intonation are well researched, an examination of visual-only input is missing. There are attempts at using visual notation for intonation teaching, notably Estebas-

Vilaplana's (2013) adaption of ToBI for teaching and learning (TL_ToBI); however, such attempts suggest combining visual notation with auditory input (Estebas-Vilaplana 2015). Our study investigates if auditory-only and visual-only input can be used to improve L2 intonation.

Furthermore, current research suggests that musical expertise positively influences SLA, including the production of suprasegmentals (Chobert & Besson 2013). Tanaka and Nakamura (2004) argue for a link between auditory memory and speakers' L2 proficiency, while Slevc and Miyake (2006) connect subjects' musical abilities to their adeptness in acquiring sound structures. Considering these studies, we predict that visual input will lead to more improvement than auditory input. Further, we expect speakers with higher musical abilities to improve intonation more substantially, regardless of the input they receive.

2. Methods

2.1. Variety

Differences in German and Swiss German varieties have been shown to influence speakers' L2 accents (Kolly 2013; Hove 2013). We thus only recorded speakers of the city variant of Bernese Swiss German (BE SwG) as a baseline and not BE SwG speakers talking in Standard German. BE SwG is known for using L*+H in prenuclear position contrary to Standard German's H*+L (Fitzpatrick-Cole 1999; Leemann 2012).

2.2. Speakers

14 speakers (8 male, 6 female) aged 23–31 participated in the study. All claimed to have intermediate English skills (CEFR B1/B2) and hold a university degree (Bachelor's degree or higher). None had knowledge in linguistics. The speakers received no financial reimbursement for their participation.

2.3. Material

One test sentence was recorded during the experiment (1).

In fact, for the first time in his life, he was not looking forward to returning to Hogwarts. (1)

The auditory input consisted of an audiobook recording of the test sentence (Rowling 2015), read by Stephen Fry, a speaker of Standard Southern British English. (2) shows a transcription of the recording.

In fact, for the first time in his life,
H* 0% | H* !H*+L H*+L 0% |
he was not looking forward to returning to Hogwarts. (2)
H*+L | H*+L | !H*+L | * L % ||

The visual input consisted of an interlinear graph drawn by the researchers based on this very recording (3).

In fact, for the first time in his life, he was not looking forward to returning to Hogwarts.

(3)

A pilot study indicated that the auditory group could not remember more than two intonation phrases (IP); we thus analyse only the first two IPs. To avoid confusion, speakers were asked to read the full sentence. The first author recorded the speakers in a quiet room on a Wiko Ridge 4G smartphone using the Android application RecForge II v1.2.0 with a 44kHz sampling rate.

2.4. Procedure

The speakers were split into two groups (auditory and visual) of seven speakers per group. Each speaker was recorded in an individual session. Sessions were conducted in German. For the pre-test reading, speakers were given a printout of the test sentence (see 1) and asked to read it using any intonation pattern they saw fit.

The groups then received group-specific training which used the same three training sentences for both groups. The two groups received instructions as follows:

(i) The auditory group received a printout of the training sentences alongside audio recordings. They were asked to listen to the reader's intonation and subsequently read each sentence while mimicking the reader's intonation. Training lasted approximately three minutes. The speakers then heard the recording of the test sentence three times. After each hearing, they were asked to read the sentence while mimicking the reader's intonation.

(ii) The visual group received a single-page instruction sheet on interpreting interlinear graphs. They then received the training sentences alongside interlinear graphs and were asked to read the sentences while trying to apply the intonation indicated. The researcher gave extensive feedback and where necessary read the sentences for the speakers. Training lasted between five and ten minutes. The speakers then received the test sentence together with an interlinear graph (see 3) and were asked to read the sentence using the intonation indicated. They were encouraged to practice beforehand.

Three recordings were made for each step. In each case, the third recording was used in our study. If this recording was unusable, the second one was taken. All recordings were transcribed in the autosegmental metrical framework (Pierrehumbert 1980). The transcriptions of pre-test and test recordings were compared with the transcription of the input (see 2) to determine whether target intonation was achieved. Only pitch accents that were realised exactly like the recording were counted as having improved. Statistical analyses were conducted in R (R 2016).

3. Results

3.1. Type of input

Figure 1 shows the relative proportion of target pitch accents produced by the visual group and the auditory group in pre-test and test reading.

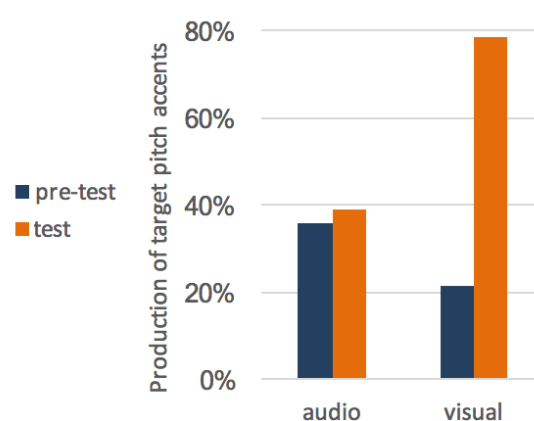


Figure 1: Target pitch accents by recording

The visual group performed significantly better in the test than in the pre-test reading (Wilcoxon signed-rank test, $p=0.026$). The auditory group showed no significant improvement (Wilcoxon signed-rank test, $p=0.564$). The difference in improvement for the visual group ($Mdn=2$,

IQR=2, i.e. on average, two pitch accents were improved) and the auditory group (Mdn=0, IQR=1, i.e., on average no improvement) is significant (Mann-Whitney U-test, $p=0.027$).

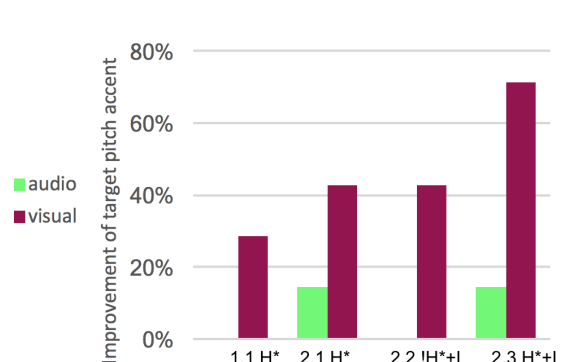


Figure 2: Improvement by type of input

Figure 2 shows the relative improvement by pitch accent for both groups. The visual group improved more for each pitch accent. However, none of the pitch accent specific improvements are significant (Fisher's exact test, $p=0.462$ for 1.1; $p=0.559$ for 2.1; $p=0.192$ for 2.2 and $p=0.103$ for 2.3).

3.2. Musical ability

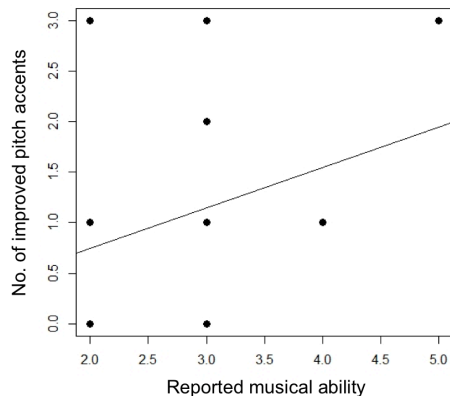


Figure 3: Association between musical ability and improvement

Figure 3 shows the relationship between speakers' musical ability and the number of pitch accents they improved. No significant relationship was found (Fisher's exact test, $p=0.772$).

4. Discussion

4.1. Type of input

Our results confirm the prediction that visual input leads to more improvement in L2 intonation than auditory input. Speakers receiving visual-only input improved

significantly more than speakers receiving auditory-only input. Only the improvements of the visual group were significant. These findings are in line with previous research (Hardison 2010; De Bot 1980, 1983; De Bot & Mailfert 1982; Weltens & De Bot 1984) which showed that audio-visual input leads to more improvement in attaining target intonation than auditory-only input. Our findings indicate that even visual input alone can lead to substantial improvement. They align closely with Catford and Pisoni (1970), who showed that conceptual understanding of how certain speech sounds are produced leads to improved production of these speech sounds; more so than imitating the sounds from a target speaker. Mimicking alone is not effective for teaching intonation but even brief instructions on interlinear graphs can lead to improvements.

Many of the visual groups' test readings were less fluent than those of the auditory group. However, we observed that those speakers who practiced more quickly improved their fluency. We also observed that speakers in the visual group who voluntarily spent more time practicing appeared to perform better in the test reading than those who practiced less. In the auditory group, practice times did not differ greatly so no similar effect was observed. Further research is needed to explore the impact of length of training.

The groups showed no significant differences for the improvement of specific pitch accents. Nevertheless, speakers in the visual group tended to perform better with regards to two-tone pitch accents (!H*+L and H*+L). We observed that two-tone pitch accents were the pitch patterns which the visual group focused on most during the training period. The H*+L pitch accent could be more foreign to produce for BE SwG speakers since their most common prenuclear pitch accent is L*+H rather than the falling pitch accent of Standard English.

4.2. Musical ability

Contrary to our expectations, we found no association between speakers' musical ability and their intonation improvement. We suggest that this result is due to our limited sample size, as a trend in Figure 3 is clearly visible.

4.3. General discussion

Our results indicate that visual-only input is more effective for improving intonation than auditory-only input. While this might be of

limited interest for classroom instructions, where teachers can easily provide audio-visual input, it may be relevant for the development of virtual tutors or low-fidelity self-study material, e.g. paper-based tutorials.

Levis and Pickering (2004), among others, criticise intonation teaching methods that focus on single sentences. They argue for using longer stretches of discourse in intonation teaching, which enables L2 speakers to understand how intonation is used in connected speech and situated contexts. Our pilot studies showed that auditory input could not be applied to utterances longer than two IPs but suggested that visual input was also applicable to longer stretches of discourse (three sentences). Further research is needed in this regard.

5. Conclusion

Our study shows that visual input is more effective for L2 intonation teaching than auditory input. Speakers' musical abilities showed to be negligible for predicting their intonation improvement. Further research is needed to analyse the use of interlinear graphs for teaching discourse-level intonation. Further research should also investigate the effects of length of training in more detail.

6. References

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